DOI: 10.20542/[placeholder]

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in M≥6.5 Earthquake Catalogs (2150 BCE – 2026 CE)

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Abstract.

We analyse a merged catalog (4,418 CSV records; 4,267 M≥6.5 events; 2150 BCE–2026 CE) using the Baiesi–Paczuski metric η with tectonic-path distance (Bird 2003). 47 global seismic series are identified (27 modern, 15 early, 5 historical candidates). Significance: permutation test (n=10,000, p<0.0001, z=−6.17); ETAS validation (FPR=0/100); FDR (45/47 at q=0.05). Fifteen early series reach p=0.007, but pre-1960 incompleteness limits interpretation. No historical series are significant (p=0.46). Largest series by event count: 1905–1910 (193 events, 43 regions, M_{max}=8.8); most spatially extensive modern series: S170 (46 events, 12 Flinn–Engdahl regions, M_{max}=9.1, 2002–2023).

Keywords: global seismicity; seismic series; earthquake clustering; tectonic distance; Baiesi–Paczuski metric; ETAS validation; false discovery rate; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that M≥7 clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global M≥7 and M≥8 rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. We test the hypothesis that multi-regional seismic series exist in a four-millennium catalog of M≥6.5 earthquakes, using an adapted η metric with tectonic distance along Bird (2003) plate boundaries. Novelty. To our knowledge, this is the first global application of nearest-neighbor clustering with tectonic-path distance across historical, early instrumental, and modern catalogs, combined with ETAS validation and FDR correction.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records from ~2150 BCE). Duplicates were merged using ±30 days and ≤50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication (±30 days, ≤50 km), the catalog contains 4,267 unique events (from 4,418 raw records). USGS ComCat contains 2,088 raw M≥6.5 events (1973–2026); after merging and deduplication, 2,041 M≥6.5 events remain (no quality_score filter). Gardner–Knopoff declustering removes ~24 aftershocks (2,017 independent, 98.8 %). quality_score 0.30–0.95 is interpretive metadata (Woessner & Wiemer 2005), not an inclusion filter. Final catalog: 4,267 M≥6.5 events (4,418 CSV rows): 2,041 modern, 2,179 early instrumental, 47 historical (fragmentary paleoseismic).

2.1.1 Catalog completeness

Maximum-curvature analysis yields M_c = 6.55. Maximum-likelihood b-value from 1,688 events above M_c:

$$b = 0.911 \pm 0.018 \quad (n = 1{,}688 \text{ events})$$

2.2 Tectonic distance and connectivity metric

Tectonic distance r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX). If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, $r_{ij} = 1.5 \times r_{GC}$ (great-circle Haversine).

$$\eta_{ij} = t_{ij} \cdot r_{ij}\text{^}1.6 \cdot 10\text{^}(-b \cdot m_i)$$

Baiesi & Paczuski [2004]; d_f=1.6; b=1.0 (code default); parent m_i only. b=1.0 follows Baiesi & Paczuski (2004); catalog empirical b=0.911±0.018 (used in Monte Carlo test). η is a relative measure; η_0 is empirical from the NN distribution.

2.3 Threshold η□ and series detection

Automatic η_0 from nearest-neighbor η distribution: KDE valley between bimodal $\log_{10}(\eta)$ modes (Zaliapin & Ben-Zion 2013); default $\eta_0 = 10^{\text{median} \log10 \eta}$. Validated against Poisson permutation null (n = 10,000).

1. Declustering via Gardner–Knopoff [1974].
2. Nearest-neighbor forest: parent i* = argmin η_{ij} subject to $\eta_{ij} < \eta_0$.
3. Global series: N ≥ 4; M ≥ 6.5; ≥ 3 Flinn–Engdahl regions.
4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

2.4 Statistical validation

Permutation test: n = 10,000, $p < 0.0001$, z = −6.17 (modern). ETAS validation (ETASCatalogGenerator): μ=0.008, K=0.08, α=1.0, c=0.005 days, p=1.1; max_trigger_distance_km=500 (links >500 km forbidden). c=0.005 days was chosen conservatively to accelerate aftershock decay in the null model; even under this strict condition, FPR=0/100. 100 synthetic catalogs (branching Poisson, seed=42); same global_series algorithm on each. FPR = fraction with ≥1 false series = 0/100; p_{ETAS} = 0.0000 (discrete test, resolution 1/101). FDR (Benjamini–Hochberg, q = 0.05): 45/47 series significant. Declustering: Gardner–Knopoff 98.8%; Zaliapin–Ben-Zion 100.0% independent.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 global seismic series: 27 modern ($p < 0.0001$), 15 early instrumental (p = 0.007; pre-1960 incompleteness caveat), 5 historical candidates (not significant, p = 0.46; 47 M≥6.5 events pre-1900). 142 cluster candidates before filtering.

ID	N	Regions	M_{max}	Period	Notes
1905–1910	193	43	8.8	1905–1910	Largest series; early instrumental
S047	53	5	8.0	1982–2024	Western Pacific subduction corridor
S170	46	12	9.1	2002–2023	Sunda belt; Sumatra 2004 (M 9.1)

S095	25	4	7.9	1989–2017	Western Pacific arc
S116	22	5	8.2	1993–2021	South Pacific multi-arc series

Table 1. Top five multi-regional series (≥ 3 Flinn–Engdahl regions). Fig. 1: figures/grl/fig01_global_map.png; Fig. 2: fig02_etas_validation.png.

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic distance lowers the η_0 threshold by ~0.3 $\log_{10}$ units compared with Euclidean distance.

4. DISCUSSION AND CONCLUSIONS

Observed global series are incompatible with Poisson and local-only ETAS nulls. Michael [2011] and Shearer & Stark [2012] tested global event rates, not multi-regional linkage structure; our η-metric is complementary. Early instrumental (1900–1972): 15 series significant at p=0.007 (z=−2.43), strengthening the conclusion that clustering is not limited to modern catalog completeness; pre-1960 quality_score<0.7 requires caution. Limitations: historical p=0.46 (47 M≥6.5 events pre-1900); p_{ETAS}=0.0000 is a discrete 100-catalog test (does not prove mechanism); correlative η-metric without depth or focal mechanisms.

1. 4,267 M≥6.5 events (4,418 CSV records) contain 47 global series; 27 modern significant at p < 0.0001.
2. ETAS validation (μ=0.008, K=0.08, α=1.0, c=0.005 d, p=1.1, 500 km; FPR=0/100) and FDR (45/47) confirm non-randomness.
3. Largest series: 1905–1910 (193 events, 43 regions, M_{max}=8.8); most spatially extensive modern: S170 (46 events, 12 regions, 2002–2023, M_{max}=9.1).
4. Tectonic distance increases η sensitivity by ~0.3 $\log_{10}$.
5. Interpretive fork: (a) if series are real — hazard implications and long-range ETAS kernels; (b) if artifacts — FDR+ETAS remains a reproducible null-test pipeline.

Future work: ΔCFS for S047, S170, S095 with depth and focal mechanisms (King et al. 1994; Stein 1999); ETAS long-range kernels; Zenodo DOI release.

DATA AND CODE AVAILABILITY

Interactive presentation: marshalkin-ux.github.io/paleoseismic-clustering/ Source code: github.com/marshalkin-ux/paleoseismic-clustering. Results: results/analysis_full_historical.json, results/etas_validation.json, results/fdr_correction_results.csv.

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Acknowledgments. The author acknowledges USGS, NOAA NGDC, and the ISC for maintaining open seismic catalogs. This work received no targeted funding.